

Trạng thái	Đã xong
Bắt đầu vào lúc	Thứ Bảy, 2 tháng 11 2024, 9:59 AM
Kết thúc lúc	Thứ Bảy, 2 tháng 11 2024, 12:49 PM
Thời gian thực hiện	2 giờ 49 phút
Điểm	11,00/11,00
Điểm	10,00 trên 10,00 (100%)

Câu hỏi 1

Đúng

Đạt điểm 1,00
trên 1,00

Write and run a program that reads two integers through the keyboard and performs simple arithmetic operations (i.e., addition, subtraction, multiplication and division) and displays the results.

Note: when performing division, you must round up the answer to 2 digits and print the answer with only 2 digits after decimal

For example:

Test	Input	Result
Test 1	5 1	5 + 1 = 6 5 - 1 = 4 5 * 1 = 5 5 / 1 = 5.00

Answer: (penalty regime: 0 %)

[Reset answer](#)

```
1 #include<iostream>
2 #include<iomanip>
3 using namespace std;
4
5 int main()
6 {
7     int a,b;
8     cin>>a>>b;
9     cout << a << " + "<<b<< " = "<<a+b<<endl;
10    cout << a << " - "<<b<< " = "<<a-b<<endl;
11    cout << a << " * "<<b<< " = "<<a*b<<endl;
12    cout << a << " / "<<b<< " = "<<fixed<<setprecision(2)<<(double)a/b<<endl;
13
14    return 0;
15 }
```


	Test	Input	Expected	Got	
✓	Test 1	5 1	5 + 1 = 6 5 - 1 = 4 5 * 1 = 5 5 / 1 = 5.00	5 + 1 = 6 5 - 1 = 4 5 * 1 = 5 5 / 1 = 5.00	✓

Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 2

Đúng

Đạt điểm 1,00
trên 1,00

Write a program that asks the user to type 5 integers (the value of 5 integers are in range [-1000000; 1000000]) and writes the average of the 5 integers. This program uses only 2 variables.

You need to complete the following code (You only write the missing code, not the whole program)

```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
    int a=0, b;
    // code that makes the program works
    cout<<fixed<<setprecision(2)<<"The average of 5 integers: "<<a/5.0;
    return 0;
}
```

For example:

Test	Input	Result
Test 1	1 2 3 4 5	The average of 5 integers: 3.00

Answer: (penalty regime: 0 %)

Reset answer

```
1  {
2      //Your code goes here
3  cin>>b;
4  a=a+b;
5  cin>>b;
6  a=a+b;
7  cin>>b;
8  a=a+b;
```

```
8 | a=a+u,  
9 | cin>>b;  
10 | a=a+b;  
11 | cin>>b;  
12 | a=a+b;  
13 | }
```



	Test	Input	Expected	Got	
✓	Test 1	1 2 3 4 5	The average of 5 integers: 3.00	The average of 5 integers: 3.00	✓

Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 3

Đúng

Đạt điểm 1,00
trên 1,00

Write and run a program that performs the following steps:

- Assigning value to a Fahrenheit temperature f . ($-300 \leq f \leq 300$)
- Calculating the equivalent Celsius temperature C using the formula: $C = (5.0/9)(f - 32)$.
- Print the answer with exact only 2 digits after decimal

Displaying the Celsius temperature C .

Please use `cout<<fixed<<setprecision(2)<< ...[your answer]... ;` to print the answer.

Example:

```
double tmp= 5.158;  
cout<<fixed<<setprecision(2)<< tmp;
```

For example:

Test	Input	Result
Test 1	255	123.89

Answer: (penalty regime: 0 %)

Reset answer

```
1 #include<iostream>  
2 #include<iomanip>  
3 using namespace std;  
4 int main()  
5 {  
6     //TODO  
7     int f;  
8     cin>>f;  
9     double c=(5.0/9)*(f-32);  
10    cout<<fixed<<setprecision(2)<<c<<endl;  
11    return 0;  
12 }
```




	Test	Input	Expected	Got	
✓	Test 1	255	123.89	123.89	✓

Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 4

Đúng

Đạt điểm 1,00
trên 1,00

Write and run a program that performs the following steps:

- Assigning value to the radius r .
- Calculating the circumference using the formula: $C = 2\pi r$.
- Displaying the circumference.

Given $\pi = 3.14$, r is an integer ($0 \leq r \leq 1000$). Just print the answer of the circumference. You don't need to round up to any digit after decimal

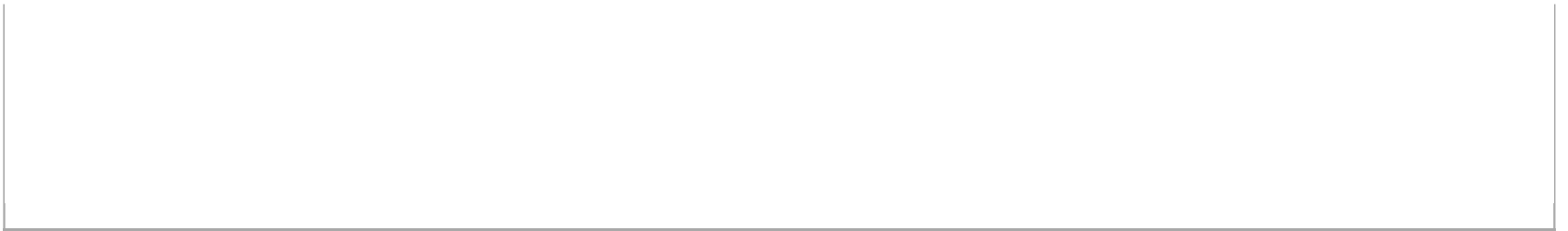
For example:

Test	Input	Result
Test 1	1	6.28

Answer: (penalty regime: 0 %)

Reset answer

```
1 #include<iostream>
2 #include<iomanip>
3 using namespace std;
4 int main()
5 {
6     int r;
7
8     cin >> r;
9     double C=2*3.14*r;
10    cout << C;
11    return 0;
12
13 }
```

	Test	Input	Expected	Got	
✓	Test 1	1	6.28	6.28	✓
✓	Test 2	0	0	0	✓
✓	Test 3	1000	6280	6280	✓
✓	Test 4	14	87.92	87.92	✓
✓	Test 5	4	25.12	25.12	✓

Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 5

Đúng

Đạt điểm 1,00
trên 1,00

Write a program that converts the number of days into years, weeks and days. Example: 1532 days = 4 years + 10 weeks + 2 days. Student needs to assign value to number of days and display the result as example. Assume a year has 365 days.

Input: one line contains number n - the number of days ($0 \leq n \leq 1000000$)

For example:

Test	Input	Result
5	1532	1532 days = 4 years + 10 weeks + 2 days.

Answer: (penalty regime: 0 %)

Reset answer

```
1 #include<iostream>
2
3 using namespace std;
4 int main()
5 {
6     int n;//days
7     cin>>n;
8     cout << n <<" days = "<< n / 365 <<" years + "<< (n % 365) / 7 <<" weeks + "<< (n % 365) % 7 <<"
9     return 0;
10
11 }
```


	Test	Input	Expected	Got	
✓	Test 1	0	0 days = 0 years + 0 weeks + 0 days.	0 days = 0 years + 0 weeks + 0 days.	✓
✓	Test 2	1000	1000 days = 2 years + 38 weeks + 4 days.	1000 days = 2 years + 38 weeks + 4 days.	✓
✓	Test 3	365	365 days = 1 years + 0 weeks + 0 days.	365 days = 1 years + 0 weeks + 0 days.	✓
✓	Test 4	100000000	100000000 days = 273972 years + 31 weeks + 3 days.	100000000 days = 273972 years + 31 weeks + 3 days.	✓
✓	5	1532	1532 days = 4 years + 10 weeks + 2 days.	1532 days = 4 years + 10 weeks + 2 days.	✓

Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 6

Đúng

Đạt điểm 1,00
trên 1,00

There are many ways to calculate the distance between A and B.

Below are 2 ways to calculate the distance between point A and point B

- Manhattan distance: $|x_A - x_B| + |y_A - y_B|$
- Euclidean distance: $\sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}$

Write and run a program that reads the coordinate of 2 points, A and B, from the

keyboard and then displays the distance between A and B in 2 ways: Manhattan distance and Euclidean distance.

When printing the distance using Euclidean distance, please round up to 2 digits after decimal by using this formula:

```
cout<<fixed<<setprecision(2)<< ...;
```

Input:

- The first line contains the coordinate of point A
- The second line contains the coordinate of point B
- Coordinate of point A and point B are integer
- $-100000 \leq x_A, y_A, x_B, y_B \leq 100000$

Hint: You can use `abs()`, `sqrt()` function

For example:

Test	Input	Result
Test 1	0 0	Manhattan distance: 2
	1 1	Euclidean distance: 1.41

Answer: (penalty regime: 0 %)

Reset answer

```
1 #include<iostream>
2 #include<iomanip>
3 #include<stdlib.h>
```

```
4  #include<math.h>
5  using namespace std;
6  int main()
7  {
8      //TODO
9      float a,b,x,y;
10     cin >>a >> b;
11     cin >>x >> y;
12     cout<<"Manhattan distance: "<< abs(a-x)+abs(b-y)<<endl;
13     cout <<"Euclidean distance: "<< fixed << setprecision(2)<<sqrt((a-x)*(a-x)+(b-y)*(b-y));
14     return 0;
15 }
```



	Test	Input	Expected	Got	
✓	Test 1	0 0 1 1	Manhattan distance: 2 Euclidean distance: 1.41	Manhattan distance: 2 Euclidean distance: 1.41	✓
✓	Test 2	14 04 14 04	Manhattan distance: 0 Euclidean distance: 0.00	Manhattan distance: 0 Euclidean distance: 0.00	✓
✓	Test 3	-5 5 1 1	Manhattan distance: 10 Euclidean distance: 7.21	Manhattan distance: 10 Euclidean distance: 7.21	✓
✓	Test 4	15 5 27 -27	Manhattan distance: 44 Euclidean distance: 34.18	Manhattan distance: 44 Euclidean distance: 34.18	✓
✓	Test 5	2020 2021 -2019 -2020	Manhattan distance: 8080 Euclidean distance: 5713.42	Manhattan distance: 8080 Euclidean distance: 5713.42	✓



Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 7

Đúng

Đạt điểm 1,00
trên 1,00

Write and run a program that reads the name, age, sex, height and weight of a student and displays with proper heading for each variable.

Given that type of name and sex is string, type of age and weight is int, type of height is float.

```
#include<iostream>
using namespace std;
int main()
{
    \\TODO
    return 0;
}
```

For example:

Test	Input	Result
Test 1	A 20 Male 1.75 85	Name: A Age: 20 Sex: Male Height: 1.75 Weight: 85

Answer: (penalty regime: 0 %)

Reset answer

```
1 #include <iostream>
2 #include <string>
3 #include <sstream>
4 using namespace std;
5
6 int main() {
7
8     char name:
```

```
8     char name;\n9     int age;\n10    float height;\n11    int weight;\n12    char sex[6];\n13    cin>>name;\n14    cin>>age;\n15    cin>>sex;\n16\n17    cin>>height;\n18    cin>> weight;\n19    cout <<"Name: "<<name<<endl;\n20    cout <<"Age: "<<age<<endl;\n21    cout <<"Sex: "<<sex<<endl;\n22    cout <<"Height: "<<height<<endl;\n23    cout <<"Weight: "<<weight<<endl;\n24    return 0;\n25 }
```



	Test	Input	Expected	Got	
✓	Test 1	A 20 Male 1.75 85	Name: A Age: 20 Sex: Male Height: 1.75 Weight: 85	Name: A Age: 20 Sex: Male Height: 1.75 Weight: 85	✓
✓	Test 2	B 18 Female 1.75 85	Name: B Age: 18 Sex: Female Height: 1.75 Weight: 85	Name: B Age: 18 Sex: Female Height: 1.75 Weight: 85	✓
✓	Test3	C 19 Male 2 85	Name: C Age: 19 Sex: Male Height: 2 Weight: 85	Name: C Age: 19 Sex: Male Height: 2 Weight: 85	✓

	Test	Input	Expected	Got	
✓	Test 4	D 21 Male 1.75 90	Name: D Age: 21 Sex: Male Height: 1.75 Weight: 90	Name: D Age: 21 Sex: Male Height: 1.75 Weight: 90	✓
✓	Test 5	E 21 Male 1.75 90	Name: E Age: 21 Sex: Male Height: 1.75 Weight: 90	Name: E Age: 21 Sex: Male Height: 1.75 Weight: 90	✓

Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 8

Đúng

Đạt điểm 1,00
trên 1,00

Write and run a program that reads an integer n ($-1000000 \leq n \leq 1000000$) from the keyboard and displays whether the number is odd or not? You MUST NOT use IF statement.

For example:

Test	Input	Result
Test 2	1	Odd

Answer: (penalty regime: 0 %)[Reset answer](#)

```
1 #include<iostream>
2 #include <string.h>
3 using namespace std;
4 int main()
5 {
6     //TODO
7     int n;
8     cin>>n;
9     string h;
10    h=(n % 2==0)? "Even" : "Odd";
11    cout << h;
12    return 0;
13 }
```



	Test	Input	Expected	Got	
✓	Test 2	1	Odd	Odd	✓
✓	Test 3	5	Odd	Odd	✓
✓	Test 4	7777777	Odd	Odd	✓
✓	Test 5	1000000	Even	Even	✓

Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 9

Đúng

Đạt điểm 1,00
trên 1,00

Given a function:

$$F(x) = ax^2 + bx + c$$

Write and run a program that performs the following steps:

- Reading the value of a, b and c from the keyboard.
- Solving $F(x) = 0$ (assume that the equation has two real distinct solutions x_1 and x_2).
- Displaying x_1 and x_2 .

Input: One line contains 3 integers a, b, c ($-100 \leq a, b, c \leq 100$)

Output: Round up the answer to 2 digits after decimal

Hint: use:

```
cout<<fixed<<setprecision(2)<< ... ;
```

For example:

Test	Input	Result
Test 1	1 2 -1	X1= 0.41 X2= -2.41

Answer: (penalty regime: 0 %)

Reset answer

```
1 #include<iostream>
2 #include<math.h>
3 #include<iomanip>
4 using namespace std;
5 int main()
6 {
7     //TODO
8     int a,b,c;
9     cin>>a>>b>>c;
```

```
10     float delta,x1,x2;
11     delta=b*b-4*a*c;
12     if(delta>0){
13         x1= (-b+sqrt(delta))/(2*a);
14         x2= (-b-sqrt(delta))/(2*a);
15     }
16
17     cout <<"X1= " << fixed<<setprecision(2)<<x1<<endl;
18     cout <<"X2= " << fixed<<setprecision(2)<<x2;
19     return 0;
20 }
```

	Test	Input	Expected	Got	
✓	Test 1	1 2 -1	X1= 0.41 X2= -2.41	X1= 0.41 X2= -2.41	✓
✓	Test 2	1 -4 3	X1= 3.00 X2= 1.00	X1= 3.00 X2= 1.00	✓
✓	Test 3	1 6 -9	X1= 1.24 X2= -7.24	X1= 1.24 X2= -7.24	✓
✓	Test 4	-5 5 7	X1= -0.78 X2= 1.78	X1= -0.78 X2= 1.78	✓
✓	Test 5	14 -4 0	X1= 0.29 X2= 0.00	X1= 0.29 X2= 0.00	✓

Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 10

Đúng

Đạt điểm 1,00
trên 1,00

Write a program that asks the user to type two integers A and B ($-100000 \leq A, B \leq 100000$) and exchange the value of A and B. The program should display the new value of A and B.

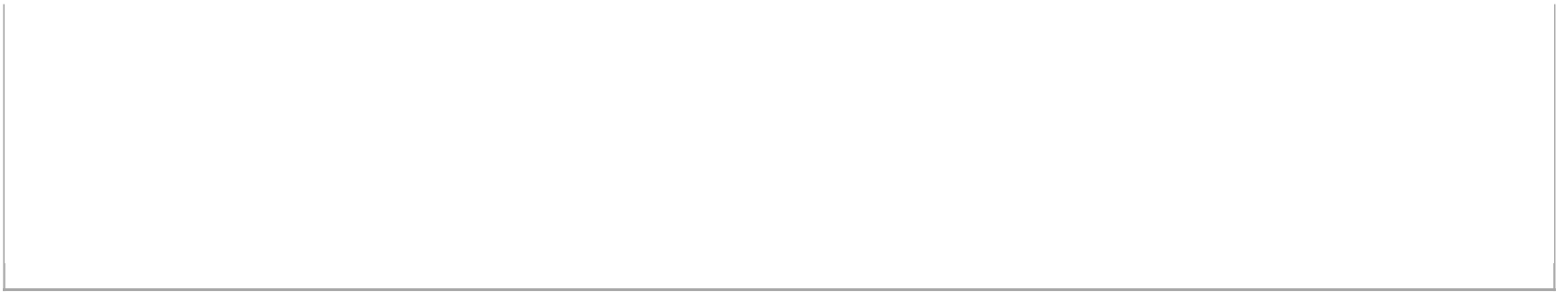
```
#include<iostream>
using namespace std;
int main()
{
    \\TODO
    cout<<A<<" "<<B;
    return 0;
}
```

For example:

Test	Input	Result
Test 1	1 2	2 1

Answer: (penalty regime: 0 %)**Reset answer**

```
1 #include<iostream>
2 using namespace std;
3 int main()
4 {
5     //TODO
6     int A,B,C;
7     cin>>A>>B;
8     C=A;
9     A=B;
10    B=C;
11    cout<<A<<" "<<B;
12    return 0;
13 }
```

	Test	Input	Expected	Got	
✓	Test 1	1 2	2 1	2 1	✓

Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.

Câu hỏi 11

Đúng

Đạt điểm 1,00
trên 1,00

Read the following C++ program for understanding. Add some suitable declarations to complete the program and run it.

```
#include<iostream>
using namespace std;
int main(){
    \\TODO
    units = 5;
    price = 12.5;
    idnumber = 12583;
    cost = price*units;
    cout <<idnumber<<" "<<units<<" "<<price<<" "<<cost<<" "<<endl;
    tax = cost*0.06;
    total = cost + tax;
    cout <<tax<<" "<<total<<endl;
    return 0;
}
```

For example:

Test	Result
test 1	12583 5 12.5 62.5 3.75 66.25

Answer: (penalty regime: 0 %)

[Reset answer](#)

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     //TODO
5     int units = 5;
6     float price = 12.5;
7     int idnumber = 12583;
```

```
7     int idnumber = 12345;  
8     float cost = price*units;  
9     cout <<idnumber<<" "<<units<<" "<<price<<" "<<cost<<" "<<endl;  
10    float tax = cost*0.06;  
11    float total = cost + tax;  
12    cout <<tax<<" "<<total<<endl;  
13    return 0;  
14 }
```



	Test	Expected	Got	
✓	test 1	12583 5 12.5 62.5 3.75 66.25	12583 5 12.5 62.5 3.75 66.25	✓

Passed all tests! ✓

► [Show/hide question author's solution \(Cpp\)](#)

Đúng

Marks for this submission: 1,00/1,00.